1. Demand and Timing

There are two inside goods owned by one monopolist and an outside option with normalized utility zero. Type r has variety preference γ_r, with $\Pr(r = H) = \lambda$ and $\Pr(r = L) = 1 - \lambda$. Consumers are myopic:

$$V_{rjt}(h) = \beta_j + \gamma_r \mathbb{1}\{j \neq h\} + \alpha p_{rjt}(h), \qquad \alpha < 0, \qquad s_{rjt}(h) = \frac{\exp V_{rjt}(h)}{1 + \sum_{k=1}^{J} \exp V_{rkt}(h)}.$$

Here p is the net transaction price. If $p = \bar{p} - d$, do not subtract d again inside utility. Discounts can be zero. No separate discount constraint is imposed in the numerical model; net prices lie in $[1, 8]$.

The initial inside history h is distributed according to q_h independently of type. Period-one prices are pooled across this initial history, even under perfect type information. After period one the firm observes the inside purchase j and can condition period-two prices on it. As in the original code, a period-one outside choice exits permanently, contributing no future profit. This is an explicit modeling assumption: allowing outside purchasers to return would require an additional continuation state.

Conditional on a single history, the share Jacobian is

$$D_{r,jk}(h) = \alpha s_{rj}(h)(\mathbb{1}\{j = k\} - s_{rk}(h)).$$

The corresponding own elasticity is $\alpha p_j(1 - s_{rj}(h))$, and the cross elasticity is $-\alpha p_k s_{rk}(h)$. For pooled initial history define

$$S_{rj} = \sum_h q_h s_{rj1}(h), \qquad D_{r,jk} = \sum_h q_h \alpha s_{rj1}(h)(\mathbb{1}\{j = k\} - s_{rk1}(h)).$$

In general $D_r \neq \alpha(\mathrm{diag}(S_r) - S_r S_r')$. Thus applying the single-logit derivative directly to averaged shares is incorrect.

2. Perfect Information and Backward Induction

Let $m_{rjt} = p_{rjt} - c_j$. Terminal continuation profits are

$$w_r(j) = \max_{p_{r2}(j)} \sum_k s_{rk2}(j)(p_{rk2}(j) - c_k).$$

The firm first solves these terminal problems and then chooses period-one prices to maximize profit per type-r initial consumer:

$$\Pi_r = \sum_j S_{rj}\{p_{rj1} - c_j + \delta w_r(j)\}.$$

Aggregate profit is $N\sum_r \omega_r \Pi_r$, where $\omega_H = \lambda$ and $\omega_L = 1 - \lambda$. In particular the discount factor belongs in $\Pi = \pi_1 + \delta\pi_2$, not $\pi_1 + \pi_2$.

The exact pooled-history first-order condition at an interior optimum is

$$0 = S_{rj} + \sum_k D_{r,kj}\{m_{rk1} + \delta w_r(k)\}.$$

For a *single* initial history, put $a_{rj} = m_{rj1} + \delta w_r(j)$. Dividing by the positive own share gives

$$0 = 1 + \alpha\left[a_{rj} - \sum_{k=1}^{J} s_{rk1}a_{rk}\right] = 1 + \alpha\left[s_{r01}a_{rj} + \sum_{k\neq j} s_{rk1}(a_{rj} - a_{rk})\right].$$

The original calculation omitted $s_{r01}a_{rj}$ even though demand has an outside option. In particular $\sum_{k\neq j,\, k\geq 1} s_{rk1} = 1 - s_{r01} - s_{rj1}$, not $1 - s_{rj1}$. Equivalently, one can sum over the outside option as well, provided its margin and continuation value are both zero.

For a single history the corrected price identity is

$$p_{rj1} = c_j - \delta w_r(j) - \frac{1}{\alpha} + \sum_k s_{rk1}\{m_{rk1} + \delta w_r(k)\}.$$

These are implicit price identities because shares depend on prices. They are not closed-form solutions. Comparing the equations for goods j and k yields

$$p_{rj1} - p_{rk1} = c_j - c_k - \delta\{w_r(j) - w_r(k)\}.$$

This relative-price identity applies only to the single-history case, not in general to the pooled-history first period used in the code.

In period two there is a single observed history j, so the corrected terminal conditions are

$$0 = 1 + \alpha\left[m_{rk2}(j) - \sum_l s_{rl2}(j)m_{rl2}(j)\right].$$

For a multiproduct monopolist with common α, interior margins are equal across inside

products, say $M_r(j)$. Therefore

$$M_r(j) = -\frac{1}{\alpha s_{r02}(j)}, \qquad p_{rk2}(j) - p_{rl2}(j) = c_k - c_l.$$

The markdown's extra term $-1/(\alpha s_{rj2}(j))$ in the relative terminal price is incorrect. It resulted from the omitted outside share. Consequently the proposed period-one formula obtained by substituting that expression is also incorrect. Binding price bounds require inequality conditions instead of these interior equalities.

3. Common Prices and Bayesian Updating

The firm knows the prior type distribution but cannot observe individual type. "Uniform" here means common across types, not a single scalar price for both goods or both periods. The posterior after an inside choice h is

$$T_h = \sum_r \omega_r S_{rh}, \qquad \lambda_2(h) = \frac{\lambda S_{Hh}}{T_h}.$$

At each history the firm reoptimizes terminal prices using this posterior:

$$p_2^*(h) \in \arg\max_p \sum_r \omega_r(h) w_r(h;p), \qquad w_r(h;p) = \sum_k s_{rk2}(h;p)(p_k - c_k).$$

There is no commitment to a terminal price that is suboptimal at the updated posterior. The total objective can be written without posterior denominators:

$$\Pi/N = \sum_r \omega_r \sum_h S_{rh}\{p_{h1} - c_h + \delta w_r(h; p_2^*(h))\}.$$

The envelope theorem then gives the particularly simple exact gradient

$$\frac{1}{N}\frac{\partial \Pi}{\partial p_{j1}} = \sum_r \omega_r \left[S_{rj} + \sum_h D_{r,hj}\{m_{h1} + \delta w_r(h; p_2^*(h))\} \right].$$

This accounts for learning: type-specific continuation profits remain inside the weighted derivative. Derivatives of optimized terminal prices drop out by the envelope theorem, not derivatives of the posterior.

For completeness the exact posterior derivative is

$$\frac{\partial \lambda_2(h)}{\partial p_{j1}} = \frac{\lambda(1-\lambda)}{T_h^2}\{D_{H,hj} S_{Lh} - S_{Hh} D_{L,hj}\}.$$

For a single initial history this simplifies, for both $h = j$ and $h \neq j$, to

$$\frac{\alpha\lambda(1-\lambda)S_{Hh}S_{Lh}}{T_h^2}(S_{Lj} - S_{Hj}).$$

The original "0 =" prefixes are erroneous: these derivatives generally are nonzero. In the expanded aggregate-profit derivative, the learning term must also carry N if the other terms do. In an isolated-price expression the alternative-product value must stay *inside* its sum over h; it cannot retain a free h outside that sum. The exact gradient above avoids both issues.

4. The Three Numerical Questions

The code uses $\beta = (2.2, 2.0)$, $c = (1.2, 1.0)$, $\alpha = -1.5$, $\delta = 0.95$, $\gamma_H = 2$, $\gamma_L = 0.1$, and $q = (1/2, 1/2)$. Consumers maximize current utility; only the firm solves a dynamic problem. Forward-looking consumers would require a different demand system.

Existing consumers' types are observed exogenously. With full information and unrestricted type-specific prices, each type's problem separates and its price is independent of its population mass. These observed-type prices and the common-price policy are useful benchmarks.

A newborn consumer is an unobserved equal-probability draw of H or L and has no purchase history. Its 50/50 type probability is an initial prior, without an individual history to update it. This is distinct from the code's original 50/50 population, which pools consumers who each have a previous inside purchase. The newborn experiment removes the initial product history: first-period utility is $\beta_j + \alpha p_{j1}$, without a switching bonus. Hence the first purchase does not reveal type. Type remains unobserved in period two; the prior $q = \Pr(L)$ persists. No fixed adoption cost or noisy classifier is introduced.

The firm compares two fixed offers. The regular menu is the dynamic policy optimized for a known H consumer. The discounted menu keeps this first-period price but replaces period-two prices by the known-L terminal optimum. All these terminal prices are verified to be below the regular menu. Holding the initial price fixed matters: a known H consumer actually receives a lower introductory price than L because their future value makes customer acquisition attractive. Thus the entire low-type dynamic policy is not a discount from the H policy.

Let Δ_r be the discounted two-period profit change from offering this period-two discount

to a newborn of true type r. The expected gain is

$$B(q) = (1-q)\Delta_H + q\Delta_L.$$

Numerically $\Delta_H = -0.04484198$ and $\Delta_L = 0.02410654$. The cost of getting type wrong is already contained in Δ_H; it is not an additional fee. The exact break-even probability is

$$q^* = -\frac{\Delta_H}{\Delta_L - \Delta_H} = 0.650369.$$

Therefore do not offer this discount when the probability of L is below approximately 65.04%; offer above this probability, with indifference at the cutoff. At the equal-probability birth prior, the offer is unprofitable. This is equivalent to not discounting when $\Pr(H) > 0.349631$.

The executable sweeps $q = 0.01, 0.02, \ldots, 0.99$ and reports the regular-menu profit, discount-menu profit, expected gain, and choice. It also reoptimizes common prices freely at each prior, which weakly dominates either fixed menu. The fixed-offer cutoff depends on the menus and does not imply a discontinuity in freely optimized prices. If instead one varied the probability a of sending a fixed offer, expected profit would be regular profit plus $aB(q)$: the optimal a would be zero or one, or any value at indifference.

See `newborn_results.md`, `newborn_prices.csv`, and `newborn_discount_probability.csv` in the results directory; the figure is `newborn_discount_cutoff.png` (also available as PDF).

The old numerical objective already jointly optimized prices in both periods. It did not use the incorrect written first-order conditions. With unconstrained terminal menus, joint optimization has optimal continuation prices at each reached posterior, so explicit backward induction should reproduce the old benchmark results. Correcting the written mathematics alone therefore need not change those numerical results.

The executable computes common prices at the known 50/50 prior, sweeps the population high-type share, and plots expected profits conditional on a grid of variety preferences. In the latter experiment it compares prices optimized for each γ against the fixed policy optimized for the original 50/50 population. A conditional expected-profit curve is not a probability density. Under the two-type distribution, profits have masses λ and $1-\lambda$ at their respective conditional expectations. A continuous profit distribution requires a specified continuous type distribution.

Run `python3 toy_model.py`; outputs are CSV tables, a numerical report, and PNG/PDF figures in `updates/toy_model/results`. The terminal and initial optimizations use bounded multistart numerical optimization; they do not prove global optimality.